

Surface Water - Is fresh water on Earth's land surface. Surface water is found in lakes, rivers, streams, and wetlands.

River system - streams and rivers move across the land and form a flowing network of water.

Watersheds -The area of land that is drained by a river.

Ground water - Water stored beneath the Earth's surface in sediment and rock formations

Water table - a level where the rocks and soil are saturated with water

Aquifer - An underground formation that contains groundwater

Porosity - is the amount of space between the particles that make up a rock.

Permeability - The ability of rock or soil to allow water to flow through it.

Recharge Zone - The area of the Earth's surface where water percolates down into the aquifer

Wells - A hole that is dug or drilled to reach groundwater

Water treatment – the process that removes elements such as mercury, arsenic, and lead, which are poisonous to humans even in low concentrations

Irrigation - is a method of providing plants with water from sources other than direct precipitation.

Dam - is a structure built across a river to control the river's flow

drip irrigation systems - deliver small amounts of water directly to plant roots by using perforated tubing.

Desalination - is the process of removing salt from salt water

Water Pollution - is the introduction of chemical, physical, or biological agents into water that degrade water quality and adversely affect the organisms that depend on the water.

Waste water - water that contains waste from homes or industry.

Artificial Eutrophication -Eutrophication caused by humans

Biomagnification – the accumulation of pollutants at successive levels of the food chain.

Primary Pollutant - A pollutant that is put directly into the air by human activity

Secondary Pollutant - form when a primary pollutant comes into contact with other primary pollutants or with naturally occurring substances such as water vapor and a chemical reaction takes place.

Zero-emission vehicles- are vehicles that have no tailpipe emissions, no emissions from gasoline, and no emission-control systems that deteriorate over time

Scrubber - is a machine that moves gases through a spray of water that dissolves many pollutants.

Smog – is produced when air pollution hangs over urban areas and reduces visibility.

Sick-building syndrome - is most common in hot places where buildings are tightly sealed to keep out the heat

Radon is one of the elements produced by the decay of uranium, a radioactive element that occurs naturally in the Earth's crust

Decibels – a measurement units to measure the intensity of sound.

Acid Precipitation - is precipitation such as rain, sleet, or snow that contains a high concentration of acids.

pH(power of hydrogen) - number is a measure of how acidic or basic a substance is.

Acid shock - the sudden influx of acidic water that causes a rapid change in the water's pH is

Climate - is the long-term prevailing weather conditions at a particular place based upon records taken

Latitude - The distance from the equator measured in degrees north or south of the equator

Wind - The movement of air within the atmosphere

Prevailing Winds - Winds that blow predominantly in one direction throughout the year

Trade winds - Belts of prevailing winds are produced in both hemispheres between 30° north and south latitude and the equator

El Nino - is the name given to the short-term (generally 6- to 18-month period), periodic change in the location of warm and cold water masses in the Pacific Ocean. During an El Niño, winds in the western Pacific Ocean, which are

usually weak, strengthen and push warm water eastward.

La Nina - the water in the eastern Pacific Ocean is cooler than usual. El Niño and La Niña are opposite phases of the El Niño–Southern Oscillation (ENSO) cycle. El Niño is the warm phase of the cycle, and La Niña is the cold phase

Pacific Decadal Oscillation - is a long-term, 20- to 30-year change in the location of warm and cold water masses in the Pacific Ocean

solar maximum - the sun emits an increased amount of ultraviolet (UV) radiation. UV radiation produces more ozone.

Ozone layer - is an area in the stratosphere where ozone is highly concentrated

Polar vortex - During the dark polar winter, strong circulating winds over Antarctica isolate cold air from surrounding warmer air

greenhouse effect – the process of heat absorption

Greenhouse gases - The gases that do absorb and radiate heat

Global Warming – The predicted increase in global temperature

Kyoto Protocol - requires developed countries to decrease emissions of carbon dioxide and other greenhouse gases

Urban - Land that is covered mainly with buildings and roads

Rural - Land that contains relatively few people and large areas of open space

Ecosystem Services - The resources that are produced by natural and artificial ecosystems

Urbanization - The movement of people from rural areas to cities

Infrastructure - is all of the things that a society builds for public use

Urban Sprawl - Rapid expansion of a city into the countryside around the city

Land-used planning - is determining in advance how land will be used—where houses, businesses, and factories will be built, where land will be protected for recreation, and so on

Geographic information system - is a computerized system for storing, manipulating, and viewing geographic data.

Open space - is land within urban areas that is set aside for scenic and recreational enjoyment.

Greenbelts - Open spaces left in their natural condition

Farmland - is land that is used to grow crops and fruit.

Rangeland - land that supports different vegetation types like grasslands, shrublands, and deserts and that is not used for farming or timber production.

Clear-cutting - is the process of removing all of the trees from an area of land.

Selective cutting - is the process of cutting and removing only middle-aged or mature trees.

Deforestation - The clearing of trees from an area without replacing them.

Reforestation - is the process by which trees are planted to re-establish trees that have been cut down in a forest land.

Wilderness - is an area in which the land and the ecosystems it supports are protected from all exploitation.

Famine - widespread starvation caused by a shortage of food

Malnutrition - is a condition that occurs when people do not consume enough Calories or do not eat a sufficient variety of foods to fulfill all of the body's needs

Drought - is a prolonged period during which rainfall is below average.

Fertile Soils - Soil that can support the growth of healthy plant

Chemical weathering - the minerals in the rock react chemically with substances such as water to form new materials

Erosion - is the wearing away of rock or soil by wind and water

Land degradation - happens when human activity or natural processes damage the land so that it can no longer support the local ecosystem.

Desertification - is the process by which land in arid or semiarid areas becomes more desert like because of human activity or climatic changes

Compost - is partly decomposed organic material.

Salinization - The accumulation of salts in the soil

Pesticides - chemicals used to kill insects, weeds, and other crop pests.

Pest - is any organism that occurs where it is not wanted or that occurs in large enough numbers to cause economic damage.

Biological pest control - is the use of living organisms to control pests

Pathogens - Organisms that cause disease

Pheromones - chemicals produced by one organism that affect the behavior of another organism, can also be used in pest control.

Integrated pest management - is a modern method of controlling pests on crops.

Genetic Engineering - the technology in which genetic material in a living cell is modified for medical or industrial use.

Overharvesting - Catching or removing from a population more organisms than the population can replace

Sustainable Agriculture - Farming that conserves natural resources and helps keep the land productive

Aquaculture - the raising of aquatic organisms for human use or consumption.

Livestock - Domesticated animals that are raised to be used on a farm or ranch or to be sold for profit

Surface mining - methods are used when ore deposits are located close to Earth's surface

Open-pit mining - is a method that is often used when large quantities of near-surface ore are mined.

Smelting - crushed ore is melted at high temperatures in furnaces to separate impurities from molten metal

Dump - Excess rock from mines is sometimes dumped into large piles

Subsidence - The sinking of regions of the ground with little or no horizontal movement

Reclamation - The process of returning land to its original or better condition after mining is completed

Fossil Fuels - are the remains of ancient organisms that changed into coal, oil, or natural gas.

Electric generator - is a machine that converts mechanical energy, or motion, into electrical energy

Turbine - is a wheel that changes the force of a moving gas or a liquid into energy that can do

work. In most power plants, water is boiled to produce the steam that turns the turbine.

Oil Reserves - are oil deposits that can be extracted profitably at current prices using current technology

Nuclear energy - the energy within the nucleus of an atom.

Nuclear Fission - collisions cause the nuclei to split

Nuclear Fusion - occurs when lightweight atomic nuclei combine to form a heavier nucleus and release tremendous amounts of energy

Renewable Energy - is energy from sources that are constantly being formed

Alternative Energy - describes energy sources that are still in development.

Tides - the movement of water in the oceans and seas caused by gravitational attraction between the sun, Earth, and moon.

Energy Efficiency - the percentage of energy put into a system that does useful work.

cogeneration - the production of two useful forms of energy from the same fuel source.

Solid waste - is any discarded solid material.

Biodegradable material - it can be broken down by biological processes.

Nonbiodegradable material - cannot be broken down by biological processes.

Active Solar Heating - Energy from the sun can be gathered by collectors and used to heat water or to heat a building

Biomass Fuel - Plant material, manure, and any other organic matter that is used as an energy source

Hydroelectric Energy - is energy produced from moving water.

Geothermal Energy - the energy from heat in the Earth's crust.

Landfill - is a permanent waste-disposal facility where wastes are put in the ground and covered each day with a layer of soil, plastic, or both

Leachate - is a liquid that has passed through compacted solid waste in a landfill.

Source Reduction - is any change in design, manufacture, purchase, or use of materials or products to reduce their amount or toxicity before they become municipal solid waste.

Recycling - is the process of reusing materials or recovering valuable materials from waste or scrap.

Photodegradable plastic - is made so that when it is left in the sun for many weeks, it becomes weak and brittle and eventually breaks into pieces.

green plastic - is made by blending the sugars in plants with a special chemical agent to make plastic

Hazardous waste - is any waste that is a risk to the health of humans or other living things

Toxicology - is the study of the harmful effects of substances on organisms.

Dose of the chemical - The amount of a harmful chemical to which a person is exposed

Response - The damage to health that results from exposure to a given dose

Epidemiology - is the study of the spread of diseases.

Risk assessment - is an estimate of the risk posed by an action or substance

Particulates - are particles in the air that are small enough to breathe into the lungs

Pesticides - are chemicals designed to kill unwanted organisms such as insects, fungi, or weeds

Pathogens - organisms that cause disease.

Host - is an organism in which a pathogen lives all or part of its life.

Vectors - Organisms, such as mosquitoes, that transmit diseases to people

Sustainability - is the condition in which human needs are met in such a way that a human population can survive indefinitely

Economics - is the study of the choices people make as they use and distribute limited resources

Economic growth - an increase in the flow of money and products within a market.

Acidification changes the balance of a soil's chemistry in several ways.

Non – point solution - comes from many different sources that are often difficult to identify

Thermal pollution can occur when power plants and other industries use water in their cooling systems and then discharge the warm water into a lake or river

Hydroelectric energy - is energy produced from moving water, is a renewable resource that accounts for about 20 percent of the world's electricity.

During deep – well injection wastes are pumped deep into the ground, where they are absorbed into a dry layer of rock below the level of groundwater.

Geothermal energy - the energy from heat in the Earth's crust.

Surface impoundment - which is basically a pond that has a sealed bottom.

Materials such as gravel that allow the flow of water are **permeable**.

Materials such as clay or granite that stop the flow of water are **impermeable**.

Point-source pollution can often be identified and traced to a source.